Development of an Anticancer agent

By Green Synthesis of Biogenic Silver Nanoparticles using agri-food byproducts as a novel reducing agent

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Introduction

Silver is considered a store of wealth. Because of its physico-chemical properties, silver is also very popular for its use in medical and chemical applications. Silver is used as an antimicrobial material and disinfectant and is considered non-harmful at appropriate concentrations. Silver nanoparticles have been shown to exemplify applications in medical imaging, filters, drug delivery, nanocomposites, cell electrodes, and antimicrobial products.[1]

Silver nanoparticles have many antibacterial, antifungal and antiviral properties. They can penetrate cell walls, modify cellular membrane structure and even cause cell decay and death. Due to its effectiveness in cellular penetration and cellular manipulation, it is an excellent candidate for anticancer agent development.[5]

While there are many options for nanoparticle synthesis, adopting biological principles can be a simple, safe, cost effective and eco-friendly way to achieve particle synthesis. This green synthesis of nanoparticles obtains much attention, mainly due to its simplistic methodology and does not require elaborate processes like maintenance of microbial culture and multiple purification steps.

This project aims to develop a potentially anti-cancer silver nanoparticle by green synthesis. In order to synthesize silver nanoparticles, a suitable reducing agent needs to be employed. **This project chose to test a few readily available food waste products in North Carolina, like Blackberry bagasse, Pecan shells, Muscadine skin and seeds, sweet potato peels, and red grape skin which are considered to have some of the highest antioxidant potentials, which would be** order to minimize waste generation and implement sustainable processes in the green synthesis of silver nanoparticles.

Objectives

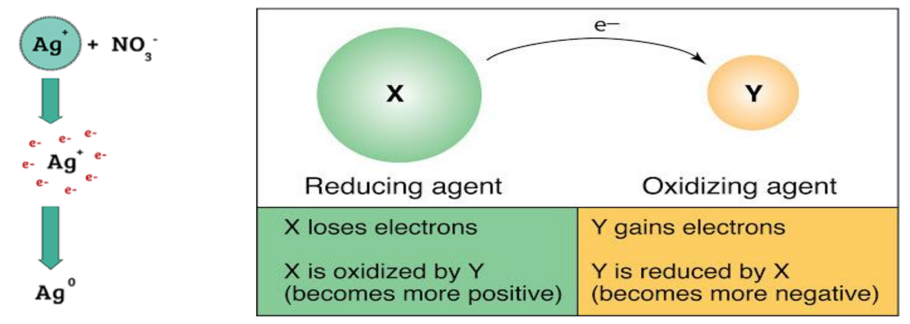
- ✓ **Determining the best Antioxidant Properties of different Agri-Food byproducts.**
- ✓ **Employing cost-effective and eco-friendly synthesis and characterization of silver nanoparticles using food waste extracts as a novel reducing agent.**

Background Research

Use of Nanotechnology

Recent advances in biotechnology and genetic engineering have made the mass production of therapeutic micro-molecules feasible. The nanoparticles became very popular in medical research because of its **effectiveness in drug delivery, cellular and antimicrobial applications** and its **ability for cellular penetration, target tissues and deep molecular targets**, as well as physico-chemical properties.

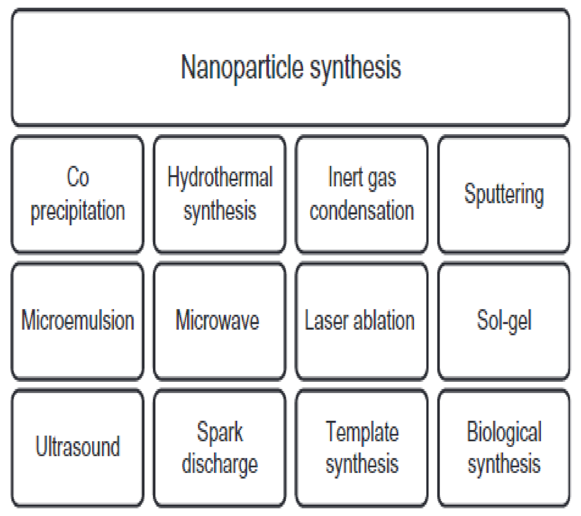
Silver Nanoparticles and its use in Cancer



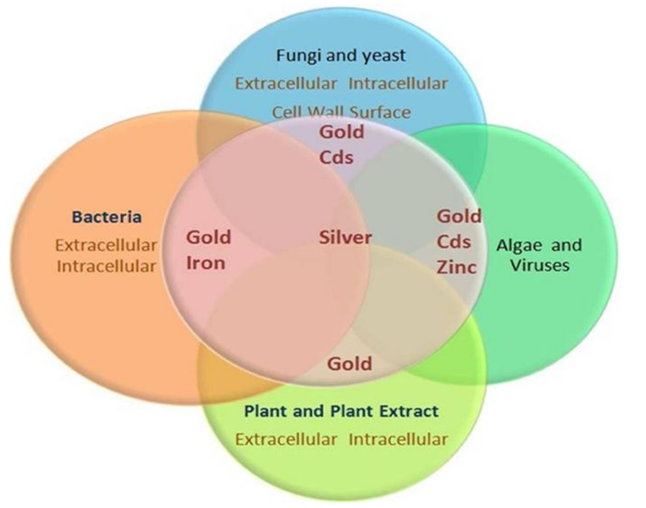
Reducing and Oxidizing agents

As they can **penetrate the tumor cells**, bind to the outer membrane of cells, act as **extracellular sustained-release drug delivery**, or be internalized into the tumor micro-environment conditions, **silver nanoparticles(AgNP) are popular in cancer research.**

- **AgNPs act as anticancer diagnostic agents** by targeting, sensing, and imaging as well as **drug delivery systems** in conjunction with **photodynamic therapy**.
- These can be regulated according to the **size of AgNPs, softness/stiffness, stability, and surface chemistry**. These factors are important in dynamic tumor and cancer metastasis controls.
- It is very important to know **specific microenvironmental conditions** that depend on the surface coating, surface hydrophobicity, hydrogen bonds, π bonds, and **treatment concentrations of AgNPs**.
- The **surface chemistry** can precisely control and tailor **Ag+ ion release conditions, pH, ionic strength, and stability.**



Nanoparticle synthesis techniques



Biological synthesis of metallic nanoparticles

Methods

Materials

Microfiber filters. Distilled water - 5 L. Methanol - 100 ml. Ice for ice bath- 1 kg 1M Ascorbic Acid - 20 ml. 1M Hydrogen Peroxide - 20 ml. 1M Potassium Permanganate - 20 ml. 2,2-diphenyl-1-picrylhydrazyl (DPPH) - 1g. Sodium Citrate - 2 g. Five readily available agri-food samples obtained locally. All chemicals are analytical grade and obtained from Sigma Chemical Co. (St. Louis, MO, USA) unless otherwise specified.

Oven dryer. Fume hood. Mortar. Pestle. Blender. Temperature probe. Vacuum Filter. Weighing Scale and weighboats. ORP Sensor with Go! Link ver 1.64. UV-vis spectrophotometer. Centrifuge with centrifugal tubes. Rotary evaporator (Rotovap). Hot Plate. Refrigerator to store samples. Computer with LoggerPro for recording the data. and Micropipette.

Readily available antioxidant-rich food items chosen for testing

Extraction of Agri-food byproducts

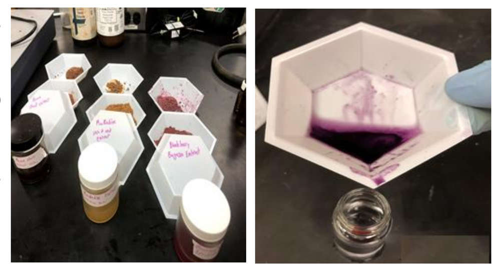
Obtained Muscadines, Blackberries, Pecan shells, Sweet Potatoes, and Red Grapes. Used mortar and pestle to grind up muscadine grapes & blackberries and cracked pecans so that muscadine skins/seeds, blackberry bagasse, pecan shells could be placed spread out in aluminum foil covered steel containers to dry in an oven at 60°C.

Antioxidant potential by ORP Sensing

Using 1M C6H8O6 (Ascorbic Acid), H2O2 (Hydrogen Peroxide), and KMnO4 (Potassium Permanganate) with ORP Sensor for calibration. Recorded data in LoggerPro to measure the Redox potentials of the solutions and find the correlation with reducing ability.

DPPH Free Radical Scavenging Assays

DPPH (2,2-diphenyl-1-picrylhydrazyl) is a π-radical that is present in a solid state, as well as in solution. Even though the p-nitro group removal has a small influence, the group removal is responsible for protecting the o-nitro groups, which led to a significant increase in reactivity. Moreover, as both phenyl groups are twisted, it has an adverse effect on conjugation stability.[2]

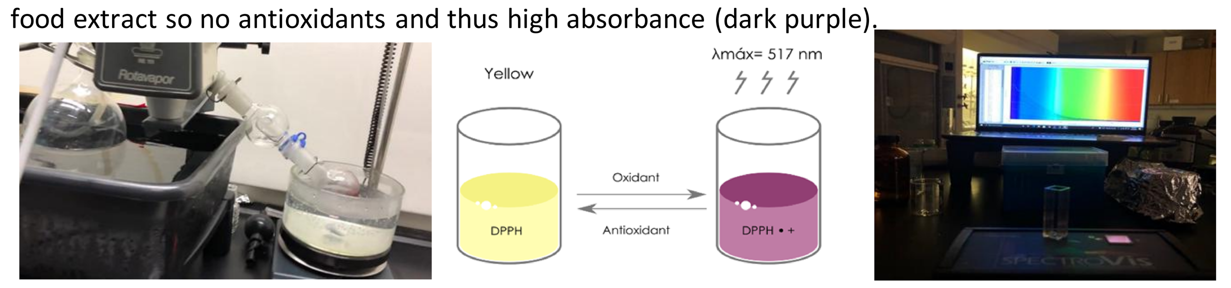


Making DPPH Assay and Measurementation

Since DPPH is insoluble in water, the extract is dried and re-dissolve it in methanol instead. A Rotovap was used to dry the extract. Added 10 mL of methanol to extract. Spun the flask on a vortex at a speed of 7.5. Then I let it sit in the refrigerator for 4 days to let the extract dissolve. The absorbance is measured at 517-520 nm and calculated according to formula:

Percentage of Antioxidant Activity of Sample = [(Absorbance of control - Absorbance of sample) / (Absorbance of sample)] *100

The **blank** will be just methanol (4-5mL). **Control** is DPPH solution in methanol w/o adding any food extract so no antioxidants and thus high absorbance (dark purple).



Extract Drying in Rotovap, DPPH Reduction and UV-vis spectroscopy of AgNPs

FTIR Spectroscopy

Synthesis of AgNPs was achieved by mixing 1 ml of extract in 9 ml of 1 mM AgNO3. Generation of AgNPs was confirmed through UV-visible spectroscopy with Vernier UV–vis spectrophotometer in the range 300-700 nm. The prepared AgNPs were added to the microfuge tubes to spin at 17,000g for 20 minutes. Decanted the solution so that only nanoparticle supernatant is left, washed nanoparticles with water. The supernatant was added to the Fourier Transform Infrared (FTIR) spectrometer for nanoparticle characterization.

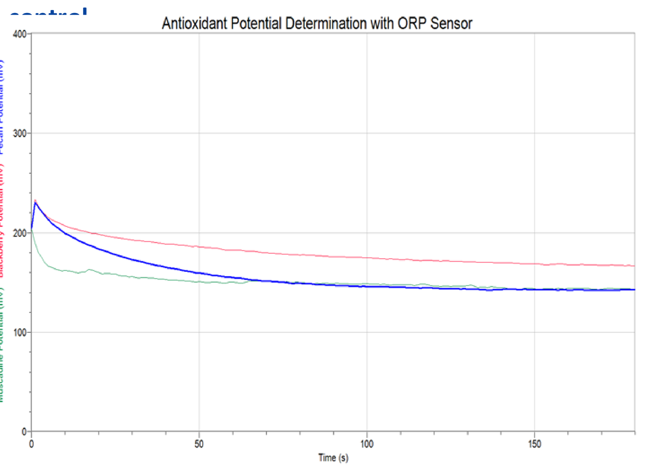
Microfuge, Extract weighing and FTIR Spectroscopy

Results and Discussion

ORP Sensing

Extract Solutions (1M)	Redox Potential (mV)
Ascorbic Acid (C6H8O6)	120.0
3% Hydrogen Peroxide (H2O2)	346.5
Potassium Permanganate (KMnO4)	578.0
Sweet Potato Peel Extract	220.0
Blackberry Bagasse Extract	170.0
Grape Skin Extract	150.0
Muscadine Skin and Seed Extract	143.5
Pecan Shell Extract	142.7

Table 1: Redox potential of food byproducts and control



Graph 1: Antioxidant potential with ORP Sensing

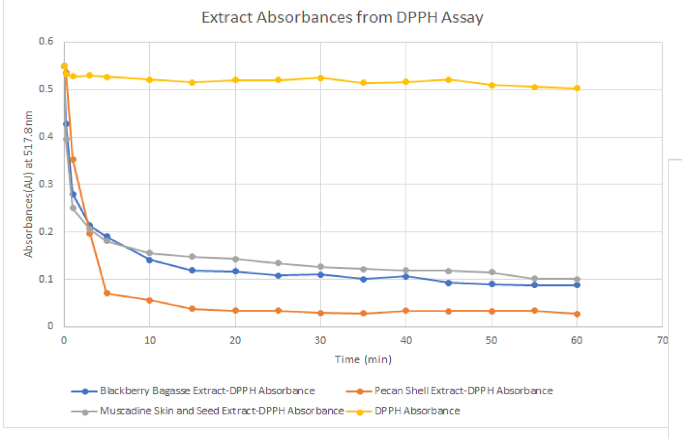
DPPH Assay

Extracts in DPPH	Antioxidant Activity (%)
Blackberry Bagasse Extract	82.903
Muscadine Skin and Seed Extract	79.920
Pecan Shell Extract	94.632

Table 2: Antioxidant Activity of Extracts in DPPH

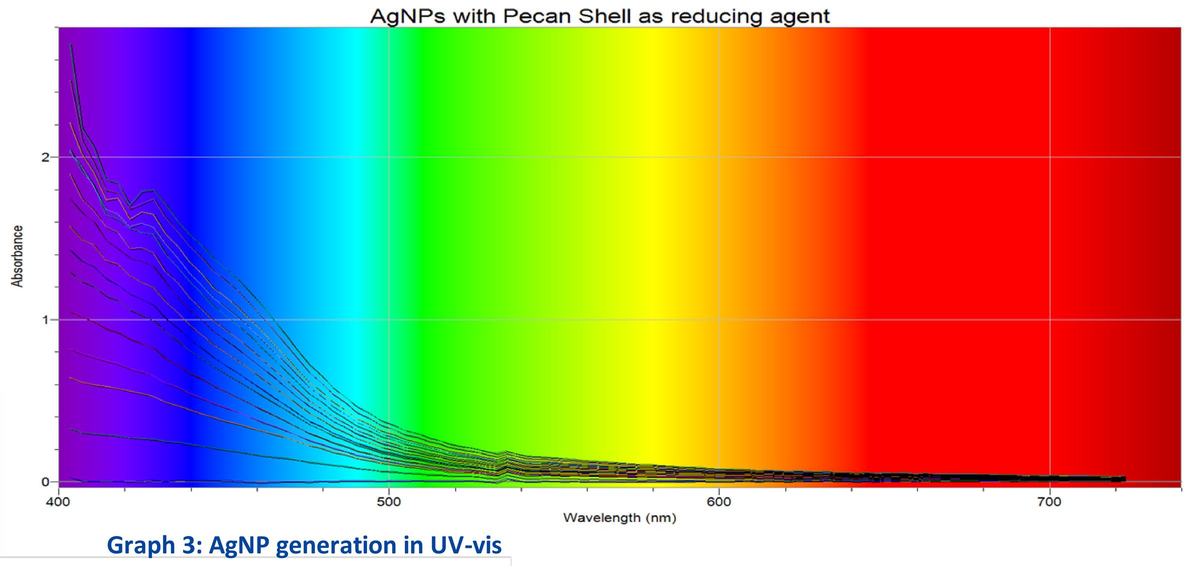
Extracts in DPPH	Half Life
Blackberry Bagasse Extract	0.319
Muscadine Skin and Seed Extract	0.325
Pecan Shell Extract	0.288

Table 3: Half-life of Extracts in DPPH

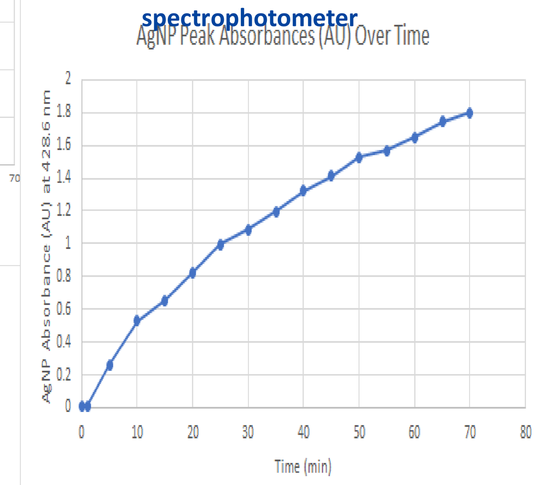


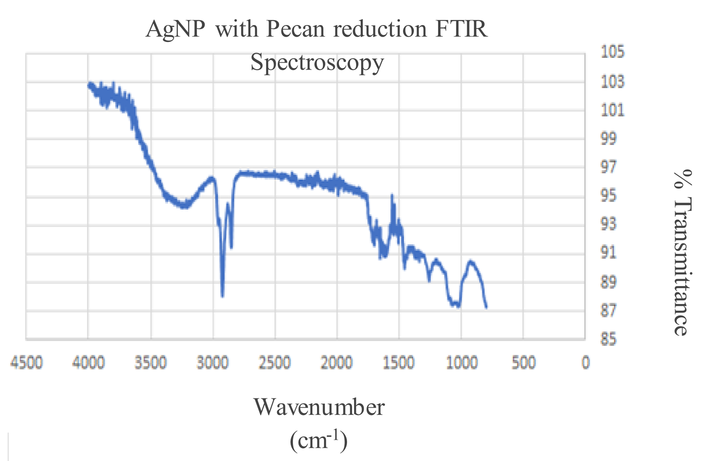
Graph 2: Extract Absorbances (AU) from DPPH Assay

Silver Nanoparticle generation and Characteristics



Graph 3: AgNP generation in UV-vis spectrophotometer





Graph 4: AgNPs peak absorbances (AU) over time   Graph 5: FTIR Spectroscopy Analysis

Discussion: Antioxidant Activity

ORP Sensor Redox Potential Analysis

The Oxidation-Reduction Potential (ORP) sensor measures a substance's ability or power to oxidize or reduce another substance. The Vernier ORP has an input range of –450 to +1100 mV at which it can measure redox potential. Readings in the positive region indicate a strong oxidizing agent, while readings in the negative region indicate a strong reducing agent (Vernier ORP Sensor User Manual).

In order to determine the best antioxidant for the purpose of reducing silver in order to synthesize a silver nanoparticle, the most negative value demonstrated by a food byproduct extract can be used. The values were an average taken from when the readings had plateaued after being submerged in the extract solutions, as shown in **Graph 1**. The **Pecan Shell extract had plateaued at the most negative value of 142.7 mV**, with the **Muscadine skin and seed extract following close behind at 143.5 mV** and the **BlackBerry bagasse extract settling at the 3rd most negative value of around 146 mV on average**. The sweet potato peel and grape skin were determined to be least effective as reducing agents due to their high redox potentials of 220 mV and 150 mV, respectively.

DPPH Assay Analysis

The DPPH Assay allows for determination of antioxidant power by analysis of the absorbance of the DPPH-extract mixture. The sample antioxidants from the extracts can react with and reduce the DPPH free radicals, resulting in the previously purple DPPH solution to become a lighter color in the presence of antioxidants. The lower the absorbance, the more the reducing power of the extract.

Conducting the DPPH assay, 3 of the food by-product powders were extracted into DMSO so that they could react with the DPPH-methanol solution. When blanked with methanol, the absorbances of DPPH mixtures over time will demonstrate if the free radicals are being reduced and by how much.

Antioxidant Activity Analysis

For measuring the Antioxidant Activity (AA), the absorbance of 3 food byproduct extracts in DPPH extract was determined using a UV-vis spectrophotometer at 517.8 nm.

The Antioxidant Activity (AA) will be calculated by the following formula:

AA (%) = [(Absorbance of control - Absorbance of sample) / (Absorbance of sample)] *100

Where AA (%) is the Antioxidant Activity in percentage.

Discussion: Physico-chemical properties of Synthesized AgNPs

UV-vis Spectrophotometry

Pecan, being demonstrated to have the most reducing power, with UV-vis spectroscopy through periodical measurements at a range of 400 nm - 700 nm. It was visibly evident that the solution had developed to become significantly more brown-ish and darker, which was reflected by the data from UV-vis spectroscopy. **The pecan shell extract solution exposed to AgNO3 ions shows a distinct absorption at around 428.6 nm (Graph 4)** which corresponds to surface plasmon resonance (SPR) of silver nanoparticles established at 420 nm in previous studies [10]. **The absorbance spectra at this peak of 428.6 nm confirmed metallic silver synthesized as nanoparticles (Graph 3). This suggests that the antioxidants present in pecan shells act as a reducing agent.**

Fourier Transform Infrared (FT-IR) Spectroscopy

The shape and size analysis was done by the FTIR spectroscopy and for the nanoparticle characterization and surface chemistry of the synthesized silver nanoparticles. Here, the **FTIR graph obtained revealed that the surface functionality of the silver nanoparticle is different from the adsorbed layer.** The FTIR spectroscopy study was carried out to recognize the biomolecules responsible for the stabilization of silver nanoparticles. This further confirms the understanding of the nanoparticle surface using the FTIR method. The nanoparticle surface charge can influence cellular uptake, toxicity, and also aggregation behavior.

Conclusion

- This research was conducted in 2 parts:
  The first part was about determining the antioxidant power of readily available food-waste byproducts.
  The second phase of the project goes on to use the effective byproducts in the biogenic green synthesis of Silver nanoparticles.
  both the objects were successfully carried out.
- The observations and data analysis on **ORP sensing and the DPPH Assay mentors proved that the hypothesis was satisfied**. above clearly demonstrate that the **hypothesis is fully accepted**. Also, the physico-chemical properties of the synthesized nanoparticles were determined by the UV-vis spectrophotometer and the FTIR spectroscopy analysis.
- **The absorbance spectra at this peak of 428.6 nm confirmed metallic silver getting synthesized as nanoparticles.** It also suggests that the antioxidants present in pecan shells could act as a good reducing agent.
- **Hence this project concludes that green-synthesis of silver nanoparticles using agri-food byproducts such as Pecan shells as a novel reducing agent** could be viably used as an alternate cost-effective and eco-friendly option in North Carolina and **exhibits its potential for development as an anticancer agent.**
- Overall, this was a fun project to execute, and I learnt what are the different ways of drug delivery and what are the challenges associated with them. I also learned the **importance of nanotechnology and how it can be used to solve some of the challenges in healthcare to help in oncology research.**

Future Goal

In this project I focused on using the antioxidant power of the readily available agri-food byproducts to reduce the silver nanoparticles, and determining the Physico-chemical properties of the synthesized Silver nanoparticles. The current goals are completed successfully. However, for determining its suitability of cancer treatment, many other analyses would be necessary.

- ❖ **Determination of the cytotoxicity effect of the silver nanoparticles against microbes to determine potential for its development as an anticancer agent.**
- ❖ **Effect of external stimulants, temperature, pressure etc. and how those affect the nanoparticles.**

Therefore, the future goal of my project is to focus on these aspects to take the results of the current study forward.

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Acknowledgements

I acknowledge my research mentors, Dr. Michael J. Bruno and Dr. Tim Anglin at NCSSM for all the help and guidance they provided in conducting my research. I also acknowledge my parents for providing me necessary books and journals. Also, I acknowledge the NCSSM Summer Research and Innovation Program (SRIP) and the Research in Chemistry program for the support to my research, and the NCSSM Chemistry laboratory for allowing me to conduct the research in the lab and use the lab equipment under supervision of my mentors.

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and Mathematics

Development of an Anti-Cancer Agent

Green Synthesis of Biogenic Silver Nanoparticles using agri-food byproducts as a novel reducing agent

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Grade 12
Research in Chemistry

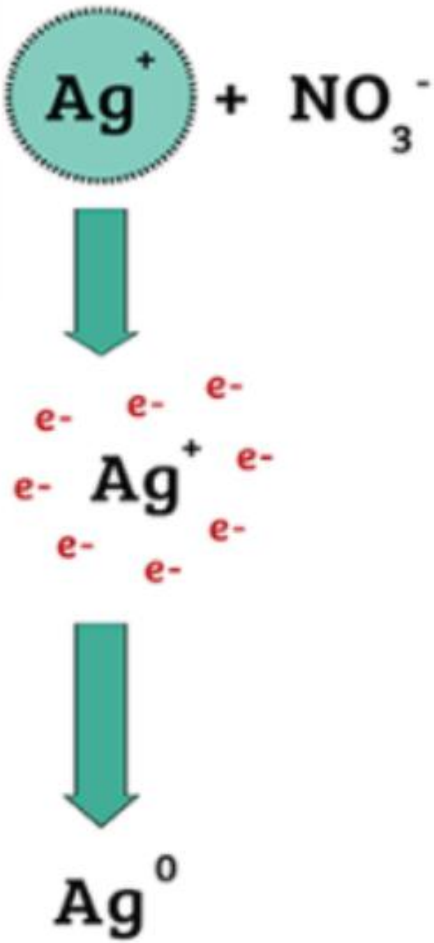
Objectives

Question:

How do different food-waste byproducts exhibiting antioxidant potential affect their use as reducing agents in the green synthesis of silver nanoparticles?

Hypothesis:

If **5 readily available food-waste byproducts in North Carolina** are tested to measure their **antioxidant potentials**, **Pecan shells or Blackberry Bagasse** are likely to exhibit the highest antioxidant potential to work as a good **reducing agent for silver nanoparticle synthesis**.



Specific Aims :

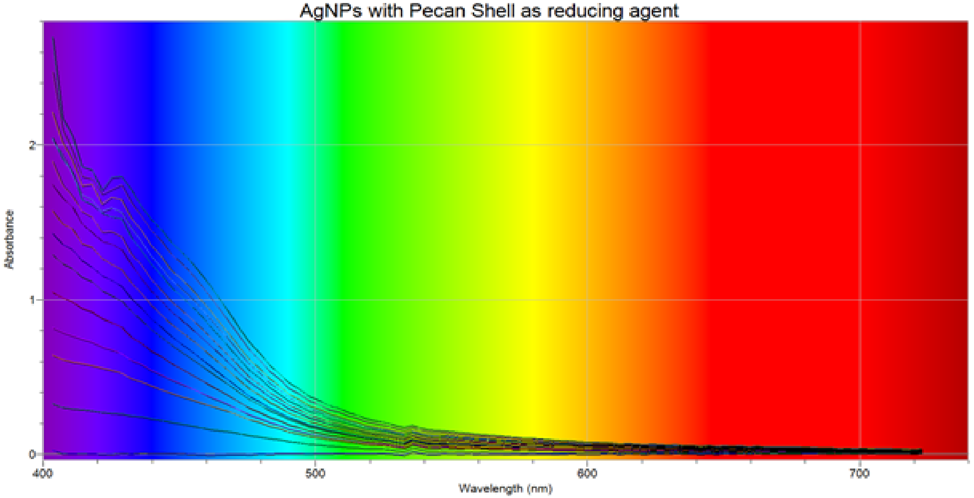
Phase 1:

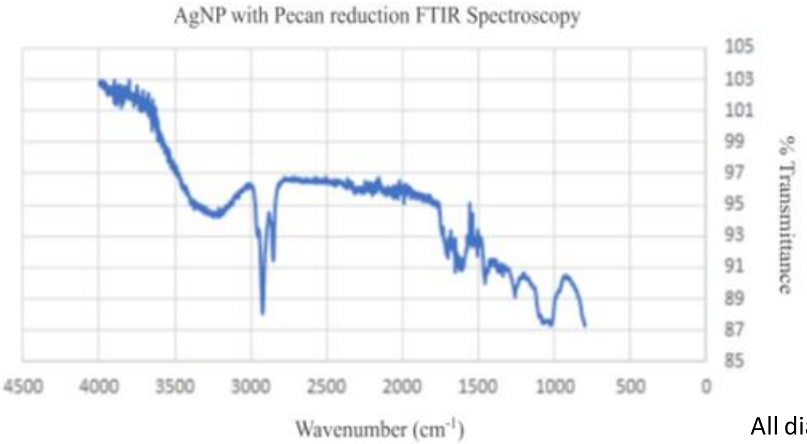
Determining the best Antioxidant Properties of different Agri-Food Byproducts

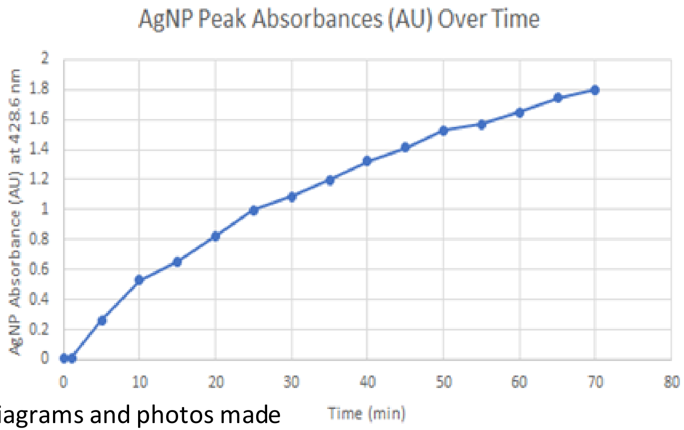
Phase 2:

Employing cost-effective and eco-friendly synthesis and characterization of silver nanoparticles using food waste extract as reducing agents

Nanoparticle Synthesis: UV-vis Spectrophotometry and FTIR







All diagrams and photos made by Hrishika Roychoudhury.

Conclusion

✅ **Objective 1**

Determining the best Antioxidant Properties of different Agri-Food Byproducts

✅ **Objective 2**

Employing cost-effective and eco-friendly synthesis and characterization of silver nanoparticles using food waste extracts as a novel reducing agent

Future Goals

a) **Determination of the cytotoxicity effect of the silver nanoparticles against microbes to determine potential for its development as an anticancer agent.**

a) **Effect of external stimulants, temperature, pressure etc. and how those affect the nanoparticles.**